

Supplementary Appendix: Unified Models, Exact Information and Covariance for Spatial Regression and CBMR

Abstract

This appendix brings together the lesion-regression manuscript `cbmr_paper.pdf`, the published multi-group CBMR model `imag.a.1057.pdf`, and the closed-form Hessian and covariance notes [3, 4]. The central distinction is between CBMR with spatially constant covariate effects (GC-CBMR) and CBMR with spatially varying covariate effects (SV-CBMR). These predictor choices are separate from the four observation models. For each model we specify the likelihood, whether data reduction is exact, and the resulting information matrix. The shared computational improvements are weighted basis products, structured inversion, and symmetric eigenvalue diagnostics. Roughness penalties, shared-covariate uncertainty, robust covariance and bootstrap inference are treated explicitly rather than conflated with likelihood simplification. Accompanying SymPy checks verify exact algebraic identities, not statistical calibration or numerical benchmarks.

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S1 Two CBMR covariate-effect specifications and notation

S1.1 The key distinction: constant effects, not constant covariate values

This appendix uses **GC-CBMR** for CBMR with *globally constant covariate effects across voxels*, and **SV-CBMR** for CBMR with *spatially varying covariate effects*. These are descriptive labels for the two parameterisations, not different observation distributions. In both cases the covariate value Z_{ir} may differ between studies. What changes is the coefficient multiplying it at voxel j :

$$\frac{\partial \log \mu_{ij}}{\partial Z_{ir}} = \begin{cases} \gamma_r, & \text{GC-CBMR: constant across } j, \\ b_r(j) = B_j^\top \beta_r, & \text{SV-CBMR: a spatial coefficient map.} \end{cases} \quad (\text{S1.1})$$

Here $\beta_r \in \mathbb{R}^P$ is the coefficient block for covariate r within the stacked SV-CBMR vector β ; group membership and other covariates are held fixed. For a one-unit change in a continuous covariate, the mean ratio is $\exp(\gamma_r)$ everywhere under GC-CBMR, but $\exp\{b_r(j)\}$ under SV-CBMR. The same ratio describes a binary covariate’s change from 0 to 1 when the predictor is linear in that covariate.

Question	GC-CBMR: constant effects	SV-CBMR: varying effects
What varies spatially?	Group baseline maps $B_j^\top \xi_g$; the moderator coefficient γ_r does not	The coefficient $b_r(j)$ of each modelled covariate
Effect of one covariate	One scalar γ_r per specified global effect	One P -coefficient basis expansion β_r per effect
Interpretation	A common multiplicative scaling of a group’s map	A location-dependent change in the spatial pattern
Main computation	Separable mean factors and grouped reductions (Section S4)	Kronecker-design block assembly, generally nonseparable means (Section S2)
Main covariate test	A global contrast of γ	A voxelwise contrast of $b_r(j)$

A spatially varying baseline does not make a covariate effect spatially varying. Also, “constant across voxels” and “shared across groups” are different restrictions: group-specific coefficients γ_{gr} can still be spatially constant within each group.

S1.2 A matched example with the same groups and one covariate

For a study covariate x_i and the same group baseline $f_g(j) = B_j^\top \xi_g$, compare

$$\eta_{ij}^{\text{GC}} = f_{g(i)}(j) + x_i \gamma_x, \quad (\text{S1.2})$$

$$\eta_{ij}^{\text{SV}} = f_{g(i)}(j) + x_i b_x(j), \quad b_x(j) = B_j^\top \beta_x. \quad (\text{S1.3})$$

GC-CBMR asks whether x_i scales the map up or down by the same ratio everywhere. SV-CBMR asks *where* its effect is positive, negative or absent. A positive γ_x raises the mean at every voxel, whereas $b_x(j)$ can have different signs at different locations.

With G groups and a basis of size P , these examples have $GP + 1$ and $(G + 1)P$ regression coefficients, respectively, before identification constraints. In the general notation below the GC-CBMR covariate matrix is $Z = x$ with a separate group-membership matrix L ; the corresponding SV-CBMR covariate matrix is $Z = [L \ x]$, whose group-indicator columns encode the baseline maps. Thus R always means the number of columns of the *specified* Z , not necessarily the same count in the two designs.

If the basis contains a constant function, $Bc_B = \mathbf{1}_N$, the restriction $\beta_x = \gamma_x c_B$ reduces this SV-CBMR example exactly to GC-CBMR. Constancy is a restriction on the *function* $B\beta_x$, not in general equality of its basis coefficients. The corresponding study mean becomes separable; merely observing similar effect estimates at a few voxels is not an exact separability condition.

S1.3 How the models fit together

The lesion manuscript [1] and the published CBMR article [2] differ in both their *predictor design* and their *observation likelihood*. These are separate choices. The former supplies the SV-CBMR specification (Design A below); the latter supplies the GC-CBMR specification (Design B below). Poisson, independent NB and clustered NB likelihoods can be differentiated under either affine predictor. The efficient aggregated-NB formulas here specifically use the grouped separable predictor.

Observation model	Data and stochastic assumption	Reduction and destination
Poisson	Independent cell counts; variance equals mean	Exact block assembly; grouped model uses both count marginals (Sections S3, S4.2)
Independent NB	Independent cell counts with variance $\mu_{ij} + \alpha_g \mu_{ij}^2$	Exact cellwise score and weights; generally no marginal-only likelihood (Section S3)
Aggregated NB	NB approximation to a sum of independent NB cells	Moment matching is approximate; differentiation of that working likelihood is exact (Section S4.3)
Clustered NB	One shared gamma multiplier per study	Exact marginalisation; correlated voxels and rank-one curvature correction (Section S4.4)

Sections S5–S7 then apply common estimation and inference machinery without changing the chosen model.

S1.4 A single notation for both papers

We retain the principal symbols of Sections 2.2–2.4 of the lesion manuscript. An observational unit i is a *subject* there and a *study* in CBMR; j always indexes voxels. The same equations do not imply the same data-generating assumptions.

Symbol	Dimension	Meaning
M, N	scalars	numbers of observational units and voxels
R, P	scalars	numbers of covariates and spatial basis functions
Y_{ij}, μ_{ij}	scalars	count and mean for unit i , voxel j
Z, B	$M \times R, N \times P$	covariate and spatial design matrices
$Z_i^\top, B_j^\top$	$1 \times R, 1 \times P$	rows of those matrices
$X = Z \otimes B$	$MN \times PR$	full design, not explicitly constructed
β	$PR \times 1$	full-model spatial regression coefficients
ξ, γ	$P \times 1, R \times 1$	separable-model spatial and global coefficients
$G, \mathcal{I}_g, M_g$	group count, set, size	$i \in \mathcal{I}_g$ iff $g(i) = g$
ξ_g	$P \times 1$	group-specific spatial coefficients
α	scalar	NB dispersion, $\text{Var}(Y_{ij}) = \mu_{ij} + \alpha \mu_{ij}^2$
C	$S \times R$	lesion-model covariate contrast matrix

S1.5 SV-CBMR (Design A): spatially varying covariate effects

Subject i varies slowest in the stacked vectors Y, μ . Define $\tilde{\beta} \in \mathbb{R}^{R \times P}$ by $\tilde{\beta}_{rp} = \beta_{(r-1)P+p}$; thus $\beta = \text{vec}_{\text{row}}(\tilde{\beta})$. The full predictor is

$$\eta_{ij} = \log \mu_{ij} = Z_i^\top \tilde{\beta} B_j = (Z_i^\top \otimes B_j^\top) \beta, \quad (\eta_{ij})_{i,j} = Z \tilde{\beta} B^\top. \quad (\text{S1.4})$$

This fixes the vectorisation convention explicitly. In particular $\beta_r = \tilde{\beta}_r^\top$ and $b_r(j) = B_j^\top \beta_r$ in (S1.1). An intercept or group-indicator column in Z supplies a spatial baseline; other columns supply spatially varying moderator effects.

S1.6 GC-CBMR (Design B): spatially constant covariate effects

The single-group separable initialisation model in the lesion manuscript is a *different parameterisation*:

$$\eta_{ij} = \eta_{Bj} + \eta_{Zi} = B_j^\top \xi + Z_i^\top \gamma, \quad \mu_{Bj} = e^{B_j^\top \xi}, \quad \mu_{Zi} = e^{Z_i^\top \gamma}, \quad \mu_{ij} = \mu_{Bj} \mu_{Zi}. \quad (\text{S1.5})$$

The published CBMR model generalises this to

$$\eta_{ij} = B_j^\top \xi_{g(i)} + Z_i^\top \gamma, \quad \mu_{ij} = \mu_{B,g(i),j} \mu_{Zi}, \quad \mu_{Bgj} = \exp(B_j^\top \xi_g). \quad (\text{S1.6})$$

Let $L_{ig} = \mathbf{1}\{g(i) = g\}$ be the $M \times G$ membership matrix, $\xi_{\text{all}} = (\xi_1^\top, \dots, \xi_G^\top)^\top$, and $\vartheta = (\xi_{\text{all}}^\top, \gamma^\top)^\top$. Then

$$\eta = \mathcal{X} \vartheta, \quad \mathcal{X} = [L \otimes B \quad Z \otimes \mathbf{1}_N], \quad \dim(\vartheta) = GP + R. \quad (\text{S1.7})$$

This restates Equations 1–5 of the published article. Set $G = 1$ to obtain (S1.5); omit the global block when there are no moderators. Group-specific global effects may be encoded by group-supported columns of Z ; in that case R counts all such columns. Separate group fits instead have separate parameter vectors and independent estimates when their data are independent.

We use $\ell(\beta, \alpha)$ for the full-model log-likelihood and $\ell(\xi, \gamma, \alpha)$ for the separable model; γ is not substituted for the full coefficient vector β .

S1.7 Translation from the published article and source notes

Published CBMR symbol	Symbol here	Reason
$X, x_j^\top$	$B, B_j^\top$	Reserve $X = Z \otimes B$ for Design A
β_g	ξ_g	Reserve β for spatially varying covariate effects
μ_g^X, μ^Z	μ_{Bg}, μ_Z	Same spatial/global mean factors
Y_{ti}, Y_{gj}	Y_i, Y_{gj}	Study total and group-voxel total
$r'_{gj}, p'_{gj}, \alpha'_g$	$\rho_g, p_{gj}, \alpha_g^{\text{eff}}$	Aggregated NB shape, probability and effective dispersion
λ_i	Λ_i	Random study multiplier, not a smoothing weight
$C (m \times S)$	$C_G (S \times G)$	S always counts tested constraints here
J	J_B	Roughness matrix; distinct from sandwich meat $\mathcal{J}$

In the source CBMR notes, (n, V, K) correspond to (M, N, P) here. Their non-spatial design G becomes Z in (S1.5), and their (u_v, e_i, θ) become $(\mu_{Bj}, \mu_{Zi}, \alpha)$. Pattern counts are denoted by G for grouped CBMR, not by P , which is reserved for basis size. Their NB shape R_p is renamed ρ_g to avoid collision with the lesion manuscript's covariate count R . C remains a contrast matrix and is not used for an information cross block. C_G and C_γ distinguish group and global-covariate contrasts from the lesion manuscript's C .

S1.8 Scope and inferential assumptions

All regression derivatives hold dispersion fixed, including when it is set to an estimate $\hat{\alpha}$ (or $\hat{\alpha}_g$). They do not propagate dispersion-estimation uncertainty. If joint observed-information inference is required, the dispersion row and column must also be included before inversion. All inverses below require an identifiable parameterisation and nonsingular stated blocks. Mean-centering covariates does not by itself establish full rank, especially when covariates are constant within groups or a basis has redundant columns. Binary data do not make a Poisson or NB working likelihood identical to a Bernoulli likelihood.

S2 SV-CBMR: exact design and Hessian assembly

For SV-CBMR, let the negative log-likelihood be $F(\beta) = \sum_{i,j} f_{ij}(\eta_{ij})$, holding all non-regression quantities fixed. Write $s_{ij} = -f'_{ij}(\eta_{ij})$ and $w_{ij} = f''_{ij}(\eta_{ij})$. With $X_i = Z_i^\top \otimes B \in \mathbb{R}^{N \times PR}$ and $W_i^{\text{obs}} = \text{diag}(w_{i1}, \dots, w_{iN})$,

$$\nabla_\beta \ell = \sum_i X_i^\top s_i, \quad (\text{S2.1})$$

$$H = -\nabla_\beta^2 \ell = \sum_i X_i^\top W_i^{\text{obs}} X_i = \sum_i (Z_i Z_i^\top) \otimes (B^\top W_i^{\text{obs}} B), \quad (\text{S2.2})$$

$$H_{(r,p),(r',p')} = \sum_i Z_{ir} Z_{ir'} \sum_j w_{ij} B_{jp} B_{jp'}. \quad (\text{S2.3})$$

Proof. The derivative of η_{ij} in $\beta_{(r,p)}$ is $Z_{ir} B_{jp}$ and every second derivative of η_{ij} is zero. Applying the chain rule to each f_{ij} therefore gives (S2.3), with no predictor-curvature term. The Kronecker mixed-product identity gives (S2.2). $\square$

This is the source notes' affine-predictor lemma in the unified notation. No independence claim about real lesion data is needed to differentiate the working likelihood. Its statistical consequences are addressed by the sandwich estimator in Section S7.

Equation (S2.2) can be evaluated subjectwise or in batches. It does not require X , an $MN \times MN$ diagonal matrix, or a $(PR) \times M \times N$ derivative intermediate. A weighted basis product $B^\top \text{diag}(w_i) B$ costs $O(NP^2)$ for a dense basis, with an $N \times P$ workspace; scatter into the R^2 coefficient blocks costs $O(R^2 P^2)$ per subject. Dense information storage still costs $O((PR)^2)$. If many subjects have exactly the same row of Z , their weights may first be summed at each voxel and the weighted basis product computed once per distinct row. No approximation is involved. The affine chain rule is shared by both specifications. For GC-CBMR (Design B) use the corresponding study rows $\mathcal{X}_i$ from (S1.7). More generally, if the study likelihood depends jointly on its log-mean vector, the same affine chain rule gives $H = \sum_i \mathcal{X}_i^\top W_i^{\text{obs}} \mathcal{X}_i$ with $W_i^{\text{obs}} = \nabla_{\eta_i}^2 (-\ell_i)$. Independent-cell models make W_i^{obs} diagonal; clustered NB has the diagonal-minus-rank-one form in (S4.29).

S3 Observation likelihoods shared by both specifications

The Poisson or NB choice does not determine whether the covariate effect is spatially constant or varying. The scalar derivatives below apply to either predictor. Projection into regression coefficients uses X_i for SV-CBMR and $\mathcal{X}_i$ for GC-CBMR.

S3.1 Scalar derivatives and Fisher weights

For one cell, suppress indices and put $\mu = e^\eta$. With parameter-free terms retained or discarded consistently, the two log-likelihoods are

$$\ell_P = Y\eta - e^\eta - \log \Gamma(Y + 1), \quad (\text{S3.1})$$

$$\ell_{\text{NB}} = \log \Gamma(Y + \alpha^{-1}) - \log \Gamma(\alpha^{-1}) - \log \Gamma(Y + 1) + Y \log \alpha + Y\eta - (Y + \alpha^{-1}) \log(1 + \alpha e^\eta). \quad (\text{S3.2})$$

For $Y \in \{0, 1\}$, the gamma terms and $Y \log \alpha$ cancel, recovering Equation 2.10 of the lesion manuscript. The exact scalar derivatives are

$$s_P = \frac{\partial \ell_P}{\partial \eta} = Y - \mu, \quad w_P = -\frac{\partial^2 \ell_P}{\partial \eta^2} = \mu, \quad (\text{S3.3})$$

$$s_{NB} = \frac{\partial \ell_{NB}}{\partial \eta} = \frac{Y - \mu}{\alpha\mu + 1}, \quad w_{NB}^{\text{obs}} = -\frac{\partial^2 \ell_{NB}}{\partial \eta^2} = \frac{\mu(Y\alpha + 1)}{(\alpha\mu + 1)^2}, \quad (\text{S3.4})$$

$$w_{NB}^F = E[w_{NB}^{\text{obs}}] = \frac{\mu}{\alpha\mu + 1}. \quad (\text{S3.5})$$

The right-hand sides of (S3.3)–(S3.5) are generated from the SymPy objects checked by `proofs/appendix.py`. The expectation uses only $E[Y] = \mu$, since the observed NB weight is affine in Y . It agrees with $\text{Var}(s_{NB}) = (\mu + \alpha\mu^2)/(1 + \alpha\mu)^2$ under the NB model. Both NB weights and the NB score have the Poisson limits as $\alpha \rightarrow 0^+$.

Thus (S2.2) with w^{obs} gives the observed information H . Replacing it by w^F gives the Fisher information

$$I(\beta) = \sum_i X_i^\top W_i X_i, \quad W_i = \text{diag}(W_{i1}, \dots, W_{iN}), \quad (\text{S3.6})$$

$$W_{ij} = \begin{cases} \mu_{ij}, & \text{Poisson,} \\ \mu_{ij}/(1 + \alpha\mu_{ij}), & \text{independent NB.} \end{cases}$$

These match the weights in the lesion manuscript's Equation 2.17. For grouped independent NB use $\alpha_{g(i)}$ in the same formulas. For Poisson $H = I$ and neither contains Y at fixed β ; evaluated at $\hat{\beta}$, both depend indirectly on the data through the fit. For NB, H and I generally differ even at a finite-sample optimum.

S3.2 The NB score is not the Poisson residual score

The same log link does *not* give the same score for the two families:

$$U_i = \nabla_{\beta} \ell_i = \begin{cases} X_i^\top (Y_i - \mu_i), & \text{Poisson,} \\ X_i^\top \text{diag}((1 + \alpha\mu_i)^{-1})(Y_i - \mu_i), & \text{independent NB.} \end{cases} \quad (\text{S3.7})$$

Vector divisions and powers here and below are elementwise. For NB Fisher scoring the exact update direction is $I(\beta)^{-1} \sum_i U_i$. Equivalently, the IRLS working response is $\eta + (Y - \mu)/\mu$ and multiplying its residual by the Fisher weights gives $(Y - \mu)/(1 + \alpha\mu)$, not $Y - \mu$. Consequently the unweighted NB residual in the lesion manuscript's update and score discussion must not be used as an NB likelihood score. This appendix retains the manuscript's notation while making that mathematical distinction explicit.

S3.3 What separability does and does not simplify

At a mean of the form $\mu_{ij} = \mu_{Zi}\mu_{Bj}$, the *full-model* Poisson information has the exact factorisation

$$I_P(\beta) = (Z^\top W_Z Z) \otimes (B^\top W_B B), \quad W_Z = \text{diag}(\mu_Z), \quad W_B = \text{diag}(\mu_B). \quad (\text{S3.8})$$

This follows by applying the mixed-product identity to $X^\top (W_Z \otimes W_B) X$. When both factors are nonsingular its inverse is the Kronecker product of their inverses. Factorisation costs $O(R^3 + P^3)$ instead of $O((PR)^3)$; materialising a dense covariance still requires $O((PR)^2)$ storage. Fixing these weights at an initial fit gives a preconditioner, not the current information of a nonseparable full-model fit. Convergence additionally requires a suitable optimisation procedure; a fixed preconditioner alone does not guarantee convergence.

For NB, even separable means do not generally yield separable Fisher weights. For two subjects with factors a, b and two voxels with factors c, d , let $W_{ij}^* = \mu_{Zi}\mu_{Bj}/(1 + \alpha\mu_{Zi}\mu_{Bj})$. Then

$$\det W^* = \frac{abcd(a-b)(-c+d)}{(aac+1)(aad+1)(abc+1)(abd+1)}. \quad (\text{S3.9})$$

This is generically nonzero for positive factors, $\alpha > 0$, $a \neq b$ and $c \neq d$. A separable weight grid has rank one, so the Poisson factorisation cannot simply be relabelled as an exact NB factorisation. The block assembly (S2.2) remains exact for either NB weight.

S4 GC-CBMR: grouped reductions and information

S4.1 Common grouped design and count summaries

Use GC-CBMR (Design B) throughout this section. Its spatially constant covariate effects yield the factorisation $\mu_{ij} = \mu_{B,g(i),j}\mu_{Zi}$. The grouped Poisson compression and aggregated-NB reparameterisation below rely on this structure and must not be transferred unchanged to unrestricted SV-CBMR. The local clustered-NB weights (S4.29), by contrast, apply to either affine predictor when projected with the appropriate design. Define $Y_{gj} = \sum_{i \in \mathcal{I}_g} Y_{ij}$, $Y_{i\cdot} = \sum_j Y_{ij}$, and $Y_g = \sum_{i \in \mathcal{I}_g} Y_{i\cdot} = \sum_j Y_{gj}$. The vector Y_g contains the group-voxel counts, not the full stacked observations. Set Z_g to the rows of Z in $\mathcal{I}_g$ and μ_{Zg} to the corresponding global mean factors.

For one group, suppress g and restrict study sums to $\mathcal{I}_g$. The cell design row for its spatial and shared global coefficients is $(B_j^\top, Z_i^\top)$. For any cellwise likelihood with observed weight w_{ij} ,

$$H_{\xi\xi} = B^\top \text{diag} \left(\left\{ \sum_i w_{ij} \right\}_j \right) B, \quad (\text{S4.1})$$

$$H_{\xi\gamma} = \sum_{i,j} w_{ij} B_j Z_i^\top, \quad (\text{S4.2})$$

$$H_{\gamma\gamma} = Z^\top \text{diag} \left(\left\{ \sum_j w_{ij} \right\}_i \right) Z. \quad (\text{S4.3})$$

Use (S3.3) or (S3.4) for the observed information and (S3.5) for independent NB Fisher information. These are $(P+R) \times (P+R)$ information matrices, not the $PR \times PR$ matrix in (S3.8). In the joint grouped fit, place each spatial block in its own position, retain its cross block with γ , and sum the global blocks over groups. Clustered NB is not cellwise independent and instead uses Section S4.4.

S4.2 Poisson: exact marginal compression, not loss of study totals

Likelihood and exact reduction. Define $E_g = \sum_j \mu_{Bgj}$, $T_g = \sum_{i \in \mathcal{I}_g} \mu_{Zi}$, $a_g = B^\top \mu_{Bg}$ and $u_g = Z_g^\top \mu_{Zg}$. Substituting the grouped predictor into the independent Poisson log-likelihood and collecting terms gives, up to data-only constants,

$$\ell_P = \sum_g Y_g^\top \log \mu_{Bg} + \sum_i Y_{i\cdot} \log \mu_{Zi} - \sum_g E_g T_g. \quad (\text{S4.4})$$

This is the computationally useful marginal expression in Equation 6 of the published article: both Y_g and $Y_{i\cdot}$ must be retained when global covariates are estimated. Computing the mean term and scores uses group-voxel arrays and study vectors rather than an $M \times N$ mean array.

There is an important distinction between this compression and fitting only $Y_{gj} \sim \text{Poisson}(\mu_{Bgj}T_g)$. Up to data-only constants,

$$\ell_{\text{full}} - \ell_{\text{totals}} = \sum_g \sum_{i \in \mathcal{I}_g} Y_i \log \left(\frac{\mu_{Zi}}{T_g} \right). \quad (\text{S4.5})$$

The missing term is the conditional multinomial allocation of each voxel's total across studies, with probabilities μ_{Zi}/T_g . Its global score is

$$\sum_g \left\{ \sum_{i \in \mathcal{I}_g} Y_i Z_i - Y_g \frac{u_g}{T_g} \right\}.$$

Thus voxel totals alone do not generally give the same covariate likelihood. If the allocation probabilities are fixed rather than estimated, that term is parameter-free for the spatial fit.

Exact information and scores. The nonzero blocks reduce to

$$\begin{aligned} H_{\xi_g \xi_g} &= T_g B^\top \text{diag}(\mu_{Bg}) B, & H_{\xi_g \gamma} &= a_g u_g^\top, \\ H_{\gamma \gamma} &= \sum_g E_g Z_g^\top \text{diag}(\mu_{Zg}) Z_g. \end{aligned} \quad (\text{S4.6})$$

The scores are

$$\nabla_{\xi_g} \ell = B^\top (Y_g - T_g \mu_{Bg}), \quad \nabla_{\gamma} \ell = \sum_i Z_i (Y_i - E_{g(i)} \mu_{Zi}).$$

For $G = 1$ these are the separable-initialisation formulas of the lesion manuscript. Differentiation is exact; faster evaluation does not change the Poisson approximation used for binary data.

S4.3 Aggregated NB: moment-matched totals and exact working curvature

Likelihood and approximation boundary. The NB calculation in `cbmr.hessian.pdf` is not (S3.2) summed over individual subjects and voxels. It is the moment-matched NB likelihood for group-voxel totals in Equations 7–9 of the published article. Define

$$s_{1g} = \sum_{i \in \mathcal{I}_g} \mu_{Zi}, \quad s_{2g} = \sum_{i \in \mathcal{I}_g} \mu_{Zi}^2, \quad A_g = \frac{s_{1g}}{\alpha_g s_{2g}}, \quad \rho_g = \frac{s_{1g}^2}{\alpha_g s_{2g}} = s_{1g} A_g. \quad (\text{S4.7})$$

Here $\alpha_g > 0$ is a group-specific extension of the lesion manuscript's global α ; setting all $\alpha_g = \alpha$ recovers a shared dispersion. The summed independent NB counts have moments

$$m_{gj} = \mathbb{E}[Y_{gj}] = \mu_{Bgj} s_{1g}, \quad (\text{S4.8})$$

$$\text{Var}(Y_{gj}) = \mu_{Bgj} s_{1g} + \alpha_g \mu_{Bgj}^2 s_{2g}. \quad (\text{S4.9})$$

Using the NB convention $\Pr(Y = y) = \Gamma(y + \rho) / [\Gamma(\rho) \Gamma(y + 1)] (1 - p)^\rho p^y$, the matching distribution has

$$p_{gj} = \frac{\mu_{Bgj}}{\mu_{Bgj} + A_g}, \quad \alpha_g^{\text{eff}} = \frac{\alpha_g s_{2g}}{s_{1g}^2} = \frac{1}{\rho_g}, \quad \text{Var}_{\text{NB}}(Y_{gj}) = m_{gj} + \alpha_g^{\text{eff}} m_{gj}^2. \quad (\text{S4.10})$$

This reduces the article's apparently voxel-specific shape r'_{gj} to a single shape ρ_g per group. The spatial factor cancels exactly. It does not turn an arbitrary NB convolution into an NB

distribution. For example, writing $s_{3g} = \sum_{i \in \mathcal{I}_g} \mu_{Zi}^3$, the third cumulant of the true sum minus that of the matched NB is

$$2\alpha_g^2 \mu_{Bgj}^3 \left(s_{3g} - \frac{s_{2g}^2}{s_{1g}} \right). \quad (\text{S4.11})$$

If every study factor in a group is equal, the independent NB summands have a common success probability and the NB convolution is exact; otherwise the higher cumulants need not match.

Suppress g within the following formulas and write $d_j = \mu_{Bj} + A$. Up to a data-only constant, the negative log-likelihood is

$$\begin{aligned} \Psi(\rho, A, \eta_B) = & - \sum_j \log \Gamma(Y_{gj} + \rho) + N \log \Gamma(\rho) - N \rho \log A \\ & + \sum_j (\rho + Y_{gj}) \log d_j - \sum_j Y_{gj} \eta_{Bj}. \end{aligned} \quad (\text{S4.12})$$

Exact information for the matched likelihood. It has diagonal log-spatial curvature

$$w_j = \frac{(\rho + Y_{gj}) \mu_{Bj} A}{d_j^2}, \quad H_{\xi_g \xi_g} = B^\top \text{diag}(w) B. \quad (\text{S4.13})$$

For global coefficients, let $\rho_r = \partial \rho / \partial \gamma_r$ and $A_r = \partial A / \partial \gamma_r$, with analogous second derivatives. Then

$$(H_{\xi_g \gamma})_{pr} = \sum_j B_{jp} \left\{ \frac{\mu_{Bj}}{d_j} \rho_r - \frac{(\rho + Y_{gj}) \mu_{Bj}}{d_j^2} A_r \right\}, \quad (\text{S4.14})$$

$$(H_{\gamma \gamma}^{(g)})_{rr'} = \Psi_{\rho\rho} \rho_r \rho_{r'} + \Psi_{\rho A} (\rho_r A_{r'} + A_r \rho_{r'}) + \Psi_{AA} A_r A_{r'} + \Psi_{\rho} \rho_{rr'} + \Psi_A A_{rr'}. \quad (\text{S4.15})$$

The global block is summed over groups. With ψ, ψ_1 denoting digamma and trigamma,

$$\Psi_\rho = - \sum_j \psi(Y_{gj} + \rho) + N \psi(\rho) - N \log A + \sum_j \log d_j, \quad (\text{S4.16})$$

$$\Psi_A = -N \rho / A + \sum_j (\rho + Y_{gj}) / d_j, \quad (\text{S4.17})$$

$$\Psi_{\rho\rho} = - \sum_j \psi_1(Y_{gj} + \rho) + N \psi_1(\rho), \quad (\text{S4.18})$$

$$\Psi_{\rho A} = -N/A + \sum_j d_j^{-1}, \quad \Psi_{AA} = N \rho / A^2 - \sum_j (\rho + Y_{gj}) / d_j^2. \quad (\text{S4.19})$$

For an explicit evaluation of the inner-map derivatives, put

$$u_1 = \sum_i \mu_{Zi} Z_i, \quad u_2 = \sum_i \mu_{Zi}^2 Z_i, \quad \mathcal{M}_1 = \sum_i \mu_{Zi} Z_i Z_i^\top, \quad \mathcal{M}_2 = \sum_i \mu_{Zi}^2 Z_i Z_i^\top,$$

with sums over $\mathcal{I}_g$, and define

$$l = u_1 / s_1, \quad h = 2u_2 / s_2, \quad \mathcal{L}_1 = \mathcal{M}_1 / s_1 - u_1 u_1^\top / s_1^2, \quad \mathcal{L}_2 = 4\mathcal{M}_2 / s_2 - 4u_2 u_2^\top / s_2^2.$$

Then

$$\nabla A = A(l - h), \quad \nabla^2 A = A\{(l - h)(l - h)^\top + \mathcal{L}_1 - \mathcal{L}_2\}, \quad (\text{S4.20})$$

$$\nabla \rho = \rho(2l - h), \quad \nabla^2 \rho = \rho\{(2l - h)(2l - h)^\top + 2\mathcal{L}_1 - \mathcal{L}_2\}. \quad (\text{S4.21})$$

These follow by differentiating $\log A = \log s_1 - \log s_2 - \log \alpha_g$ and $\log \rho = 2 \log s_1 - \log s_2 - \log \alpha_g$. They retain the inner-map curvature terms in (S4.15). The spatial weight tends to $s_1 \mu_{Bj}$ as

$\alpha_g \rightarrow 0^+$. The reparameterisation is exact for this aggregated likelihood; it does not make the moment-matching approximation exact for arbitrary independent NB sums. The derivatives hold the *original* α_g fixed, not α_g^{eff} . If instead effective dispersion is independently parameterised and held fixed, ρ_g is constant and $A_g = \rho_g/s_{1g}$; its derivatives must be recomputed for that convention. Mixing these two conditioning choices changes the Hessian.

S4.4 Clustered NB: exact study marginalisation and rank-one curvature

Likelihood and dependence. The clustered extension places a shared gamma effect within each subject: $\Lambda_i \sim \text{Gamma}(\nu_g, \text{rate } \nu_g)$, $\nu_g = 1/\alpha_g$, and, conditionally on Λ_i , Y_{ij} are independent Poisson with mean $\Lambda_i \mu_{Bgj} \mu_{Zi}$ for $i \in \mathcal{I}_g$. Independent Λ_i across studies give

$$\text{Cov}(Y_i) = \text{diag}(\mu_i) + \alpha_g \mu_i \mu_i^\top, \quad \text{Cov}(Y_i, Y_{i'}) = 0 \quad (i \neq i'). \quad (\text{S4.22})$$

This includes both the diagonal variance and the off-diagonal covariance in Equation 10 of the published article; the latter is absent under independent NB.

Let $t_i = \sum_j \mu_{ij} = E_g \mu_{Zi}$. Exact marginalisation can be understood as a total-count model followed by spatial allocation:

$$Y_{i\cdot} \sim \text{NB}(\text{mean } t_i, \text{shape } \nu_g), \quad Y_i | Y_{i\cdot} \sim \text{Multinomial} \left(Y_{i\cdot}, \left\{ \frac{\mu_{Bgj}}{E_g} \right\}_j \right). \quad (\text{S4.23})$$

Multiplying these two mass functions cancels the total factorial and the allocation powers of t_i , giving the joint gamma–Poisson mass function. Unlike aggregated NB, this reduction is exact. Collecting its terms over studies gives Equation 11 of the article, up to data-only terms:

$$\begin{aligned} \ell_g = & |\mathcal{I}_g| \nu_g \log \nu_g - |\mathcal{I}_g| \log \Gamma(\nu_g) + \sum_{i \in \mathcal{I}_g} \log \Gamma(Y_{i\cdot} + \nu_g) \\ & - \sum_{i \in \mathcal{I}_g} (Y_{i\cdot} + \nu_g) \log(E_g \mu_{Zi} + \nu_g) + \sum_j Y_{gj} \eta_{Bgj} + \sum_{i \in \mathcal{I}_g} Y_{i\cdot} \eta_{Zi}, \end{aligned} \quad (\text{S4.24})$$

where $E_g = \sum_j \mu_{Bgj}$. Write $d_i = E_g \mu_{Zi} + \nu_g$, $a_g = B^\top \mu_{Bg}$ and

$$g'_g = \sum_{i \in \mathcal{I}_g} \frac{(Y_{i\cdot} + \nu_g) \mu_{Zi}}{d_i}, \quad g''_g = - \sum_{i \in \mathcal{I}_g} \frac{(Y_{i\cdot} + \nu_g) \mu_{Zi}^2}{d_i^2}.$$

Exact observed information. Then

$$\nabla_{\eta_{Bg}}^2 (-\ell_g) = g'_g \text{diag}(\mu_{Bg}) + g''_g \mu_{Bg} \mu_{Bg}^\top, \quad (\text{S4.25})$$

$$H_{\xi_g \xi_g} = g'_g B^\top \text{diag}(\mu_{Bg}) B + g''_g a_g a_g^\top, \quad (\text{S4.26})$$

$$H_{\xi_g \gamma} = a_g \left\{ \sum_{i \in \mathcal{I}_g} \frac{(Y_{i\cdot} + \nu_g) \nu_g \mu_{Zi}}{d_i^2} Z_i^\top \right\}, \quad (\text{S4.27})$$

$$H_{\gamma \gamma}^{(g)} = \sum_{i \in \mathcal{I}_g} \frac{(Y_{i\cdot} + \nu_g) \nu_g E_g \mu_{Zi}}{d_i^2} Z_i Z_i^\top. \quad (\text{S4.28})$$

The rank-one term is a negative correction since $g''_g < 0$. Only subject totals enter these Hessian weights: the terms containing the voxel marginals are linear in the coefficients and differentiate away. The result follows from differentiating a scalar function of $E_g = \sum_j e^{\eta_{Bgj}}$ twice. Thus the basis projection still needs only a weighted Gram matrix and one outer product, not an $N \times N$ matrix.

Expected information and fixed-dispersion curvature. For a single study expressed in its full log-mean vector η_i , the observed information before basis projection is

$$W_i^{\text{obs}} = \frac{Y_i + \nu_g}{t_i + \nu_g} \left\{ \text{diag}(\mu_i) - \frac{\mu_i \mu_i^\top}{t_i + \nu_g} \right\}, \quad W_i^{\text{F}} = \text{diag}(\mu_i) - \frac{\mu_i \mu_i^\top}{t_i + \nu_g}. \quad (\text{S4.29})$$

The expectation uses $E[Y_i] = t_i$. These weights are *matrices*, not the diagonal independent-NB weights. For any real vector v ,

$$(t_i + \nu_g) v^\top W_i^{\text{F}} v = \nu_g \sum_j \mu_{ij} v_j^2 + \sum_{j < k} \mu_{ij} \mu_{ik} (v_j - v_k)^2. \quad (\text{S4.30})$$

Consequently the conditional regression objective is convex for fixed $\nu_g > 0$ and an affine log predictor. The same is true for independent Poisson and NB with fixed dispersion. This is not a claim of joint convexity when dispersion is also estimated, nor of convexity of the moment-matched NB composition in (ξ_g, γ) .

S5 Roughness penalties and the fitted objective

S5.1 An explicit quadratic-penalty convention

The published article introduces a spatial roughness penalty in Section 2.1.1 and Figure 2. To carry that improvement through the Hessian calculation, let $J_B = J_B^\top \succeq 0$ be a fixed $P \times P$ roughness matrix and $\lambda_g \geq 0$ fixed smoothing weights. We use the explicit convention

$$\mathcal{P}(\vartheta) = \frac{1}{2} \sum_g \lambda_g \xi_g^\top J_B \xi_g = \frac{1}{2} \vartheta^\top K \vartheta, \quad K = \text{blkdiag}(\lambda_1 J_B, \dots, \lambda_G J_B, 0_{R \times R}). \quad (\text{S5.1})$$

The factor 1/2 is a convention: a penalty written without it has twice the gradient and Hessian for the same numerical λ_g . No particular roughness operator or smoothing weight is inferred from the PDF; its detailed construction is referred to Supplementary Appendix A.1 of the article, which is not included in the supplied PDF.

For the minimised objective $F_{\text{pen}} = -\ell + \mathcal{P}$,

$$\nabla F_{\text{pen}} = -\nabla \ell + K \vartheta, \quad H_{\text{pen}} = H + K. \quad (\text{S5.2})$$

Thus the closed-form information receives a fixed block-diagonal addition; the shared-global border and cross-group structural zeros are preserved. For SV-CBMR (Design A) use $K = \text{blkdiag}(\lambda_1 J_B, \dots, \lambda_R J_B)$ on the R spatial maps. A roughness penalty need not penalise constant directions and therefore need not remove an intercept-identification problem.

S5.2 Curvature is not automatically a sampling covariance

For an unpenalised likelihood, model-based covariance uses inverse information. For a penalised estimator, inverse penalised curvature is not generally its frequentist sampling covariance. Linearising the penalised estimating equation gives

$$V_{\text{pen}} \approx (H + K)^{-1} \mathcal{J} (H + K)^{-\top}, \quad (\text{S5.3})$$

where $\mathcal{J}$ is the variance estimate of the *unpenalised* data score, not of a score with the penalty repeatedly added per study. Under a correctly specified model replace the bread H and score variance by I to obtain $(I + K)^{-1} I (I + K)^{-1}$. In particular,

$$(I + K)^{-1} - (I + K)^{-1} I (I + K)^{-1} = (I + K)^{-1} K (I + K)^{-1}. \quad (\text{S5.4})$$

This identity distinguishes penalised curvature from sampling variation; neither expression by itself accounts for smoothing bias or data-driven smoothing-parameter uncertainty. Alternating dispersion and regression updates, as in the article’s Algorithm 1, does not remove these distinctions. A final inference step must state whether it uses likelihood information, penalised curvature, or a penalised estimating-equation sandwich.

S6 From information to covariance

The following linear-algebra identities apply to a symmetric nonsingular information matrix, whether H is observed information or I is the Fisher information. They do not change which statistical estimator has been chosen.

S6.1 Structural zeros are determined by the design

For the full independent-cell model, (S2.3) shows that the (r, r') coefficient block vanishes if no subject loads both covariates: $Z_{ir}Z_{ir'} = 0$ for every i . More generally, connect two covariate columns when they co-occur in any row of Z , and take connected components. Different components have zero cross blocks, so after reordering,

$$H = \text{blkdiag}(H_1, \dots, H_{K_{\text{comp}}}), \quad H^{-1} = \text{blkdiag}(H_1^{-1}, \dots, H_{K_{\text{comp}}}^{-1}). \quad (\text{S6.1})$$

Here K_{comp} counts connected components, while K in Section S5 is a penalty matrix. The conclusion follows from the factor $Z_{ir}Z_{ir'}$ and block multiplication. Two columns can have a zero direct cross block while remaining in the same component through a third column; that does not license separate inversion. Cell-means indicators for disjoint groups separate, whereas a shared intercept or overlapping continuous covariates generally connect the spatial columns.

The group-specific ξ_g in Section S4 likewise have no cross-group spatial Hessian terms. Shared γ coefficients usually couple the groups through a border; the full covariance then need not be block diagonal.

S6.2 Schur inversion when global coefficients form a border

Let $d = GP$ be the number of grouped spatial coefficients; the number of global coefficients is R . To keep the scalar intensity total E_g distinct from an information block, write

$$H = \begin{pmatrix} \mathcal{D} & \mathcal{F} \\ \mathcal{F}^\top & \mathcal{E} \end{pmatrix}, \quad \mathcal{D} = \text{blkdiag}(D_1, \dots, D_{K_{\text{comp}}}), \quad \mathcal{F} \in \mathbb{R}^{d \times R}, \quad \mathcal{S} = \mathcal{E} - \mathcal{F}^\top \mathcal{D}^{-1} \mathcal{F}. \quad (\text{S6.2})$$

If $\mathcal{D}$ and $\mathcal{S}$ are nonsingular,

$$H^{-1} = \begin{pmatrix} \mathcal{D}^{-1} + \mathcal{D}^{-1} \mathcal{F} \mathcal{S}^{-1} \mathcal{F}^\top \mathcal{D}^{-1} & -\mathcal{D}^{-1} \mathcal{F} \mathcal{S}^{-1} \\ -\mathcal{S}^{-1} \mathcal{F}^\top \mathcal{D}^{-1} & \mathcal{S}^{-1} \end{pmatrix}. \quad (\text{S6.3})$$

Multiplying by H proves the identity using $\mathcal{E} = \mathcal{S} + \mathcal{F}^\top \mathcal{D}^{-1} \mathcal{F}$. With $\mathcal{F} = 0$ it reduces to independent block inversion. Use solves with D_k and $\mathcal{S}$ rather than explicitly forming intermediate inverses whenever only contrasts or selected entries are required. Writing $d_k = \dim D_k$, factorisation costs $O(\sum_k d_k^3 + R^3)$; solving for $\mathcal{D}^{-1} \mathcal{F}$ costs $O(\sum_k d_k^2 R)$, forming the Schur complement costs $O(dR^2)$, and forming the dense spatial covariance correction costs $O(d^2 R)$. An invertible H can have a singular leading block $\mathcal{D}$, so invertibility of H alone is not sufficient for this particular Schur formula.

S6.3 Shared covariates induce cross-group covariance

For grouped CBMR take $D_g = H_{\xi_g \xi_g}$ and $F_g = H_{\xi_g \gamma}$, so $\mathcal{F}$ stacks the F_g vertically. Under the observed-information covariance choice $V_\vartheta = H^{-1}$,

$$(V_\vartheta)_{\gamma\gamma} = \mathcal{S}^{-1}, \quad (\text{S6.4})$$

$$(V_\vartheta)_{\xi_g \xi_h} = \mathbf{1}\{g = h\} D_g^{-1} + D_g^{-1} F_g \mathcal{S}^{-1} F_h^\top D_h^{-1}. \quad (\text{S6.5})$$

Thus a block-diagonal *spatial information block* does not imply independent spatial estimates after estimating shared γ . For global-covariate inference, $\mathcal{E}^{-1}$ would treat the spatial coefficients as known; $\mathcal{S}^{-1}$ accounts for their estimation. The same block identities hold using I instead of H , or for penalised curvature after replacing $\mathcal{D}$ by its penalised version, but the latter is not automatically the sampling covariance (S5.3).

This matters directly for Equation 14 of the published article: the small dimension of γ does not ensure nonsingular or well-conditioned profile information. A global covariate confounded with group intercepts can make $\mathcal{S}$ singular regardless of the magnitude of its coefficient. Independently fitted groups have zero cross-covariance only when their data and fitted parameter vectors are genuinely separate.

S6.4 Condition number from symmetric eigenvalues

If $H = Q\Lambda Q^\top$ with orthogonal Q , then $H^\top H = Q\Lambda^2 Q^\top$. Consequently,

$$\kappa_2(H) = \frac{\max_a |\lambda_a(H)|}{\min_a |\lambda_a(H)|}. \quad (\text{S6.6})$$

Positive definiteness is not needed for this identity. For a singular matrix the finite inverse and finite condition number are unavailable. For (S6.1), the characteristic polynomial is the product of the block characteristic polynomials, hence

$$\kappa_2(H) = \frac{\max_k \max_a |\lambda_a(H_k)|}{\min_k \min_a |\lambda_a(H_k)|}.$$

This is not in general $\max_k \kappa_2(H_k)$. For a bordered matrix, the eigenvalues of $\mathcal{D}$ and $\mathcal{S}$ are *not* the eigenvalues of H ; use the full symmetric eigenproblem. All these equalities are exact algebraically; numerical eigensolvers and inverses still incur rounding error.

An appropriate computation uses structural blocks when available, otherwise Schur inversion when its hypotheses hold, otherwise Cholesky for a positive-definite matrix or a general factorisation for a nonsingular indefinite matrix. The latter is an algebraic inverse, not evidence of a valid positive covariance estimate. Failed factorisations or singularity must be reported rather than replaced by an apparently valid covariance.

The source covariance note also considers LAPACK's `pocon`, which estimates κ_1 , not κ_2 . Norm equivalence gives $n^{-1}\kappa_1(H) \leq \kappa_2(H) \leq n\kappa_1(H)$ for an $n \times n$ matrix; it neither makes the norms equal nor bounds the estimator's error by itself. A symmetric eigensolver targets the required 2-norm quantity. The source notes' timing and memory measurements are empirical results, not consequences of the symbolic proofs and not benchmarks of this appendix.

S7 Model-based and robust voxelwise inference

S7.1 SV-CBMR: voxelwise covariate-effect contrasts

All quantities in this section are evaluated at the fitted parameters. The lesion manuscript's model-based choice is $V_\beta = \widehat{\text{Cov}}(\hat{\beta}) = I(\hat{\beta})^{-1}$. An observed-information alternative uses $H(\hat{\beta})^{-1}$;

they agree for Poisson, not generally for NB. For $\mathcal{T}_j = C \otimes B_j^\top$, the voxelwise log-contrast and covariance are

$$\hat{\eta}_{Cj} = \mathcal{T}_j \hat{\beta}, \quad V_{Cj} = \mathcal{T}_j V_\beta \mathcal{T}_j^\top. \quad (\text{S7.1})$$

For S linearly independent contrast rows with nonsingular V_{Cj} , the quadratic statistic $\hat{\eta}_{Cj}^\top V_{Cj}^{-1} \hat{\eta}_{Cj}$ has the usual asymptotic χ_S^2 null law under the statistical regularity conditions. When $S = 1$, its signed square root is the Wald statistic. The degrees of freedom must be reduced if the tested constraints are redundant. A contrast need not have row sum one; difference contrasts naturally have row sum zero.

S7.2 GC-CBMR: group-map contrasts and global covariate tests

For the published CBMR predictor, let $C_G \in \mathbb{R}^{S \times G}$ contain group contrasts and use the full covariance V_θ appropriate to the fitting/inference method. Define

$$\mathcal{T}_{Gj} = [C_G \otimes B_j^\top \quad 0_{S \times R}], \quad \hat{\delta}_j = \mathcal{T}_{Gj} \hat{\vartheta} - \delta_{0j}, \quad V_{\delta j} = \mathcal{T}_{Gj} V_\theta \mathcal{T}_{Gj}^\top. \quad (\text{S7.2})$$

The reference value δ_{0j} specifies the null, such as a known homogeneous log-intensity or a zero log difference. The statistic $\hat{\delta}_j^\top V_{\delta j}^{-1} \hat{\delta}_j$ restates Equation 12 of the article with dimensionally consistent notation. For one group, a log-map covariance across voxels is $B V_{\xi_g \xi_g} B^\top$; the voxel j variance is $B_j^\top V_{\xi_g \xi_g} B_j$, with no transpose reversal.

For the log-intensity difference between groups g and h ,

$$\text{Var}(\hat{\eta}_{Bgj} - \hat{\eta}_{Bhj}) = B_j^\top (V_{\xi_g \xi_g} + V_{\xi_h \xi_h} - V_{\xi_g \xi_h} - V_{\xi_h \xi_g}) B_j. \quad (\text{S7.3})$$

The cross terms from Section S6.3 must be retained. No balancing of M_g and M_h is required algebraically; unequal group sizes enter the likelihood and its information.

If intensity rather than log-intensity is reported, the Jacobian for $\mu_{Bg} = \exp(B\xi_g)$ is $\text{diag}(\mu_{Bg})B$. For the vector of group intensities at voxel j , the delta-method covariance is

$$V_{\mu,j} \approx \text{diag}(\hat{\mu}_{B1j}, \dots, \hat{\mu}_{BGj}) V_{\eta,j} \text{diag}(\hat{\mu}_{B1j}, \dots, \hat{\mu}_{BGj}).$$

Here $V_{\eta,j} = (I_G \otimes B_j^\top) V_{\xi_{\text{all}} \xi_{\text{all}}} (I_G \otimes B_j)$. The Jacobian identity is exact; the covariance transformation is a first-order approximation. Log contrasts avoid that additional approximation. For global contrasts, use $C_\gamma \hat{\gamma}$ and $C_\gamma V_{\gamma\gamma} C_\gamma^\top$, with $V_{\gamma\gamma} = \mathcal{S}^{-1}$ under the observed-information choice, not the inverse global diagonal block.

S7.3 Independent-unit sandwich covariance

For independent subjects, allowing arbitrary within-subject dependence, let U_i be the family-specific score in (S3.7) and define

$$\mathcal{A} = - \sum_i \frac{\partial U_i}{\partial \beta^\top}, \quad \mathcal{J} = \sum_i U_i U_i^\top, \quad V_{\text{robust}} = \mathcal{A}^{-1} \mathcal{J} \mathcal{A}^{-\top}. \quad (\text{S7.4})$$

For likelihood scores the empirical sensitivity $\mathcal{A}$ is H . Under a correctly specified conditional mean, the expected sensitivity is I for the independent NB working score, so using I as bread is an asymptotically justified alternative. For Poisson they coincide exactly. For a general misspecified estimating equation, its own derivative must be used; NB Fisher weights combined with unweighted Poisson scores are not the matching likelihood sandwich. These formulas treat dispersion as fixed. If dispersion is jointly estimated and its uncertainty is to be propagated robustly, stack its estimating equation with the regression score before forming bread and meat.

The subject-clustered choice of $\mathcal{J}$ is essential: replacing it by a sum of separate voxel score outer products discards within-subject cross terms. For a particular contrast one can avoid storing the full meat:

$$\mathcal{T}_j V_{\text{robust}} \mathcal{T}_j^\top = \sum_i v_{ij} v_{ij}^\top, \quad v_{ij} = \mathcal{T}_j \mathcal{A}^{-1} U_i. \quad (\text{S7.5})$$

This follows by distributing the matrix product over the subject sum. The independent-cell scores above must not be reused unchanged for the aggregated or clustered likelihoods of Section S4. For a grouped penalised fit use its corresponding scores, covariance (S5.3), and projection $\mathcal{T}_{Gj}$.

S7.4 Identification and the limits of algebraic improvements

In the separable model, if $Bc_B = \mathbf{1}_N$ and $Zc_Z = \mathbf{1}_M$, then $(\xi, \gamma) \mapsto (\xi + tc_B, \gamma - tc_Z)$ leaves every mean unchanged. A constraint or removal of a redundant intercept is therefore needed before using (S6.3). Similarly, adding an intercept column to a basis that already spans the constant function can create rank deficiency. Neither a large sample size nor a well-behaved condition number alone establishes valid asymptotic inference; identifiability, independent clusters and adequate information in each tested direction are separate requirements.

The article reports an empirical threshold of roughly 200 foci per group for its parametric-inference experiments. This is not a universal theorem, and exact Hessians do not establish a new validity threshold. Removing an autodifferentiation intermediate or replacing an SVD changes computational cost, not the information in a sparse dataset. Likewise, adding a small diagonal term solely to make inversion succeed changes the inferential object and is not a silent substitute for a specified penalty.

S7.5 Bootstrap remains a distinct inferential procedure

The parametric-bootstrap approach in Section 2.1.4 of the published article addresses calibration when its Wald approximation is unreliable. It is not made unnecessary by a faster Hessian. A model-consistent bootstrap fits the stated null, simulates under its observation/dependence model with the original groups and covariates, refits with the same constraints and penalty-selection rule, and recomputes the chosen statistic. Shared gamma effects must be preserved for clustered NB; a matched aggregate-NB sampling law and an independent-cell NB sampling law are not interchangeable. This prescription is distinct from a conditional spatial-homogeneity randomisation scheme and its assumptions.

Write B_{boot} for the number of replicates, reserving B for the spatial basis. A common nonzero Monte Carlo tail estimate is

$$\hat{p} = \frac{1 + \sum_{b=1}^{B_{\text{boot}}} \mathbf{1}\{T_b^* \geq T_{\text{obs}}\}}{B_{\text{boot}} + 1}.$$

The tail and statistic must match the tested alternative. A fitted-null parametric bootstrap is not thereby an exact finite-sample test. The article's generalised-Pareto tail extrapolation introduces an additional approximation that needs its own diagnostics. Neither bootstrap validity, tail calibration nor the article's empirical threshold is established by the symbolic checks here.

S8 Exact symbolic verification and reproducibility

The repository runs the four original CBMR proof modules together with `proofs/appendix.py` and `proofs/published_cbmr.py`. Each claim is an expression difference reduced by SymPy; acceptance requires the literal scalar zero or a zero matrix, not a numerical tolerance. The generated report from the current run is

Symbolic suite	Claims	Closed
Poisson Hessian	13	yes
Negative Binomial Hessian	27	yes
Clustered Negative Binomial Hessian	10	yes
Covariance strategies	9	yes
Integrated appendix	35	yes
Published CBMR integration	26	yes
Total	120	yes

The original modules check the Poisson and clustered-NB coefficient Hessians, the aggregated-NB reparameterisation and derivative components, and the structural, spectral and inverse identities. The appendix adds the lesion manuscript’s independent NB score, observed and expected weights, binary likelihood reduction, Poisson limits, coefficient ordering, generic affine and separable Hessian assembly, Poisson Kronecker factorisation, the distinction between constant and varying covariate effects and its constant-function restriction, the NB nonseparability identity, clustered curvature, and projected contrast/sandwich identities. The published-CBMR suite checks grouped Poisson compression and conditional allocation, the NB matching moments, clustered marginal factorisation and curvature, fixed quadratic penalties, and shared-global covariance corrections. The generated report counts algebraic claims; it is not a count of independently proved statistical theorems.

S8.1 Where the published model is improved

Published article	Integrated calculation	Boundary of the result
Equations 1–5	Group design (S1.7) and basis-product assembly	Same predictor, no full stacked design allocation
Equation 6	Both count marginals in (S4.4)	Voxel totals alone omit the study-allocation score
Equations 7–9	ρ_g, A_g and (S4.15)	Exact derivatives of a moment-matched likelihood
Equations 10–11	(S4.23) and (S4.29)	Exact marginalisation, including within-study dependence
Figure 2, Section 2.1.1	Fixed penalty gradient, curvature and sandwich	Smoothing bias and tuning uncertainty remain separate
Equations 12–14	Group projections and Schur covariance	Shared global effects couple spatial estimates
Section 2.1.4	Conditioning diagnostics and bootstrap boundaries	Faster algebra does not certify sparse-data Wald inference

S8.2 What is and is not machine checked

Scalar identities hold for symbolic variables with the stated positivity and reality assumptions. Explicit matrix checks use small fixed dimensions with arbitrary symbolic entries: the appendix assembly uses $M = N = P = R = 2$, while the original likelihood checks use three voxels and two subjects. Generic scalar functions check the affine chain-rule composition; the aggregated-NB components are checked separately rather than simplifying its entire composite Hessian in one pass. The original Schur check and the appendix Kronecker inverse check use formal matrix symbols of arbitrary conformable dimensions under invertibility assumptions.

The published-CBMR suite also uses small grouped and study-level examples with symbolic counts and means (and a fixed allocation-score design). Its penalty differentiation holds the roughness matrix and smoothing weights fixed; the penalised covariance-difference identity is checked in the scalar case. The dimension-independent arguments are given in the text; fixed-size

symbolic instances are not proofs over all dimensions. The generated scalar formulas and report are tied directly to the checked objects, but the handwritten exposition is not mechanically certified. SymPy is a computer algebra system, not a proof-assistant kernel. These checks also do not verify NiMARE implementation code, floating-point stability, asymptotic distributions, model adequacy, or empirical performance.

S8.3 Lean formalisation: a checked subset, not complete coverage

The repository also contains a pinned Lean/Mathlib development in `CbmrProofs/`. It formalises finite-index predictor and stacked-design identities, real-variable likelihood derivatives, vector quadratic-penalty derivatives, NB algebra and structured covariance identities. These proofs are checked by Lean rather than by symbolic simplification in SymPy. Many finite-sum and matrix statements use arbitrary finite index types instead of fixed matrix sizes.

The full appendix has not yet been converted to Lean. In particular, full derivative-to-Hessian assembly, the remaining aggregated-NB derivatives, probabilistic expectation and mixture results, spectral condition-number theory and statistical asymptotics have outstanding proof obligations. The section-by-section inventory is `docs/lean_coverage.md`. The SymPy table above must not be read as a Lean theorem count.

`make lean` compiles the included proofs and audits their transitive axiom dependencies. Admitted proofs and additional axioms are rejected; the accepted foundations are `propext`, `Classical.choice` and `Quot.sound`. Positivity, differentiability and invertibility assumptions remain explicit in theorem statements. Unproved obligations are recorded in the inventory, not asserted as axioms.

The equation-level mapping in `docs/lean_coverage.json` covers all 54 displayed mathematical blocks and additional prose derivations. It records the referenced theorems and any missing steps. `make lean-complete` additionally rejects partial or pending coverage; a successful `make lean` alone does not establish completeness. The audit output and coverage report are saved in `.lake/lean-audit.log` and `.lake/lean-coverage-report.json`.

S8.4 Rebuilding

From the repository root:

```
pip install -r requirements.txt
make appendix
```

This verifies all claims, regenerates the LaTeX fragments and compiles `docs/cbmr_appendix.pdf` twice using `pdflatex`. Any failed claim stops the build. `make proofs` runs only the symbolic checks. CI performs the same checks and rejects stale generated fragments; it does not currently build PDFs. The editable source is `docs/cbmr_appendix.tex`; the reference PDFs are left unchanged.

References

- [1] *Scalable Spatial Model for Brain MRI Lesion Mask Data*. Supplied working manuscript, `docs/cbmr_paper.pdf`, Sections 2.2–2.4. Source of the principal notation in this appendix.
- [2] Yu, Y., Hill-Bowen, L. D., Riedel, M. C., Bottenhorn, K., Laird, A. R., and Nichols, T. E. (2025). *CBMR: Coordinate-based meta-regression for group and covariate inference*. *Imaging Neuroscience*, 3. [doi:10.1162/IMAG.a.1057](https://doi.org/10.1162/IMAG.a.1057). Supplied as `docs/imag.a.1057.pdf`; this appendix adapts the mathematics rather than reproducing the article’s exposition.
- [3] *Closed-Form Observed Information for Coordinate-Based Meta-Regression*. Repository derivation note, `docs/cbmr_hessian.tex`.
- [4] *From the Hessian to the Covariance: Generalising the CBMR standard-error step*. Repository derivation note, `docs/cbmr_covariance.tex`.